Error Propagation in Multi-Step Agentic AI Workflows for Research Automation

Abstract

Agentic artificial intelligence (AI) systems are increasingly being developed to perform complex tasks through sequences of planning, reasoning, information retrieval, tool use, memory, verification, and generation. Research automation is a particularly promising application because agentic systems can assist with literature discovery, evidence extraction, synthesis, data analysis, and report generation. However, the reliability of a multi-step workflow depends not only on the capabilities of the underlying large language model (LLM), but also on the correctness of intermediate outputs. An error introduced during an early stage can become an input to later stages and subsequently propagate through the workflow. This paper examines error propagation in multi-step agentic AI workflows used for research automation. It reviews the development of reasoning-and-acting agents, tool-using language models, reflective agents, and agent evaluation frameworks. The paper identifies major error sources, including planning errors, retrieval failures, source-selection errors, reasoning mistakes, tool-use failures, memory contamination, and verification failures. It further examines sequential propagation, error amplification, error masking, and feedback-loop failures. Existing approaches such as external grounding, tool use, reflection, self-correction, provenance tracking, and human oversight are discussed as mechanisms for improving reliability. Finally, the paper identifies research gaps involving workflow-level error metrics, causal error attribution, independent verification, reproducibility, and systematic benchmarks for autonomous research agents. The analysis suggests that reliable research automation requires validation throughout the workflow rather than relying exclusively on final-output evaluation.

Keywords: Agentic AI, Large Language Models, Research Automation, Error Propagation, AI Agents, Reliability, Tool Use, Citation Verification, Multi-Step Workflows, Human-in-the-Loop

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1. Introduction

Large language models (LLMs) have evolved from systems primarily designed for text generation into components of increasingly autonomous AI agents. Modern agentic systems can reason about objectives, decompose tasks, interact with external tools, retrieve information, maintain intermediate state, and execute multiple actions before producing a final answer. ReAct, for example, combines reasoning and acting so that a language model can interleave reasoning with actions performed against external environments or information sources [1]. This approach demonstrated improvements on question answering, fact verification, and interactive decision-making tasks while addressing some problems associated with purely internal reasoning.

Research automation is an important application of this development. A research-oriented AI agent may be asked to formulate a search strategy, locate scholarly publications, determine their relevance, extract evidence, compare findings, synthesize results, and generate a report. Such a system could potentially reduce the time required for literature review and other research activities. However, research tasks are particularly sensitive to errors because research conclusions depend on chains of evidence. A mistake in one stage can therefore affect multiple downstream stages.

The central challenge is **error propagation**. In a multi-step agentic workflow, intermediate outputs are often used as inputs to later operations. If an early retrieval agent selects an irrelevant paper, for example, a later summarization agent may generate an apparently accurate summary of that irrelevant source. A synthesis agent may then incorporate the summary into a broader argument, and a final writing agent may present the argument with a convincing citation. The final output can therefore appear scholarly even though its foundation is defective.

The problem is not restricted to hallucinated references. A real source can also be misinterpreted, incorrectly summarized, or cited for a claim that it does not support. Consequently, assessing the reliability of agentic research systems requires evaluation of the complete workflow rather than only the final generated text.

This paper examines the following questions:

1. What are the major sources of error in multi-step agentic research workflows?
2. Through what mechanisms can errors propagate between workflow stages?
3. What techniques can detect, prevent, or reduce propagated errors?
4. What research gaps remain before agentic systems can reliably automate research tasks?

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2. Background and Related Work

2.1 Agentic AI and Multi-Step Reasoning

Agentic AI systems differ from conventional question-answering systems because they can perform sequences of actions toward a goal. Instead of directly mapping a prompt to a response, an agent can follow a cycle such as:

Goal → Reason → Act → Observe → Update → Act → Final Output

ReAct is an important example of this architecture. Yao et al. proposed interleaving reasoning traces with actions, allowing the model to use external information while maintaining and updating an action plan [1]. Their experiments included question answering, fact verification, and interactive environments. The authors reported that interaction with external information sources helped address hallucination and error propagation in reasoning-only approaches.

This architecture is particularly relevant to research automation because research naturally involves repeated interactions with external information. A research agent may search databases, inspect papers, execute calculations, retrieve additional information, and revise its plan.

However, every additional step creates another opportunity for failure. Consequently, agentic architectures can simultaneously increase capability and increase the complexity of reliability analysis.

2.2 Tool-Augmented Language Models

External tools provide language models with capabilities that are difficult to obtain through language generation alone. Tool-using approaches allow models to access calculators, APIs, search systems, databases, code execution environments, and other external resources.

Tool use can reduce certain types of hallucination because information can be obtained from external systems rather than generated solely from the model's parameters. However, tool integration introduces new failure modes. An agent may select an inappropriate tool, formulate an incorrect query, provide invalid parameters, or misunderstand the tool's output.

For research automation, this means that retrieval accuracy is determined by more than the quality of the search database. It also depends on how the agent formulates queries, selects sources, interprets results, and transfers information to later stages.

2.3 Reflection and Self-Correction

Reflection-based approaches attempt to make agents learn from their previous attempts. Reflexion, for example, uses verbal feedback and an episodic memory mechanism rather than updating model parameters. The agent reflects on feedback from previous attempts and stores the resulting information for subsequent decision-making [2]. The authors reported improvements across sequential decision-making, coding, and language-reasoning tasks.

Self-correction is potentially valuable for research workflows. If an agent recognizes that a retrieved source is irrelevant or that a calculation is inconsistent, it can revise the intermediate result.

Nevertheless, reflection does not guarantee correctness. If an agent receives incomplete feedback or incorrectly diagnoses the cause of failure, it may preserve the same error or produce a new one. Thus, self-correction should be distinguished from independent verification.

2.4 Evaluation of LLM Agents

Traditional language-model benchmarks often evaluate individual responses. Agentic systems, however, need to perform sequences of dependent actions. AgentBench was developed to evaluate LLMs as agents across multiple interactive environments. The study evaluated 27 API-based and open-source LLMs across eight environments and identified long-term reasoning, decision-making, and instruction-following difficulties as important obstacles to reliable agent behavior [3].

This type of evaluation is important because an agent can perform well on isolated subtasks but fail when those subtasks are connected into a long workflow. Research automation therefore requires evaluation at both the component level and the workflow level.

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3. Error Sources in Research-Oriented Agentic Workflows

A research automation workflow can be represented as:

Research Question → Planning → Literature Search → Source Selection → Evidence Extraction → Analysis → Synthesis → Verification → Report Generation

Errors can enter at any stage.

3.1 Planning Errors

Planning is the first major source of potential failure. An agent may misunderstand the research question or divide it into inappropriate subtasks.

For example, a question about the reliability of AI research agents might require separate analysis of retrieval accuracy, reasoning reliability, citation correctness, tool reliability, and human oversight. If the planning agent focuses only on model accuracy, later stages may never investigate the other dimensions.

Planning errors are particularly significant because they occur early in the workflow and can affect many downstream operations.

3.2 Information Retrieval Errors

Research agents often depend on search systems and external databases. Retrieval failures can include:

- failure to retrieve relevant papers;
- retrieval of irrelevant papers;
- poor search-query formulation;
- duplicate sources;
- outdated sources;
- incorrect interpretation of search results.

A retrieval error can be especially damaging when later agents assume that every selected source is relevant.

3.3 Source-Selection Errors

Even when a search system returns relevant publications, an agent must decide which sources should be used. It may select a paper because its title appears relevant while overlooking limitations in its methodology or population.

Source-selection errors can consequently transform a retrieval problem into an evidence problem.

3.4 Evidence-Extraction Errors

A paper may be genuine and relevant while the extracted evidence is incorrect. An agent can:

- select the wrong passage;
- omit important limitations;
- confuse authors' hypotheses with findings;
- misread numerical results;

- treat an example as a general conclusion.

This type of error is particularly dangerous because the resulting summary may still sound plausible.

3.5 Reasoning Errors

The agent may correctly retrieve evidence but make an invalid inference from it. A common example is interpreting correlation as causation.

Reasoning errors can propagate because later stages generally treat previous conclusions as inputs rather than independently re-deriving them.

3.6 Citation Errors

Citation errors include incorrect authors, publication years, titles, journals, page numbers, or persistent identifiers. More importantly, a citation can point to a genuine publication while failing to support the specific claim for which it is used.

Therefore, citation verification requires at least two separate questions:

Does the cited source exist?

and

Does the source actually support the claim?

3.7 Tool-Use Errors

External tools increase an agent's capabilities but also introduce additional failure points. An agent may choose the wrong tool, provide incorrect arguments, misread the response, or fail to recognize an error returned by the tool.

3.8 Memory Errors

Long-running research agents may maintain memory containing intermediate conclusions, source summaries, or previous decisions. If incorrect information enters this memory, it can influence multiple later steps.

This creates the possibility of **persistent error propagation**, in which an error is not merely passed to the next step but remains available to many future operations.

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4. Mechanisms of Error Propagation

4.1 Sequential Propagation

The simplest mechanism is sequential propagation:

Error → Incorrect Intermediate Output → Downstream Dependency → Final Error

For example:

Wrong source → Wrong summary → Wrong synthesis → Unsupported conclusion

The number of opportunities for propagation increases as the workflow becomes longer.

A simplified model can express the probability of an entirely successful sequence as:

$$P(\text{success}) = \prod_{i=1}^{n} p_i$$

where p_i represents the probability that stage i produces a correct output. This model assumes independent stages and is therefore only a conceptual approximation. Real agentic workflows contain dependencies between stages, making the actual reliability relationship more complex.

4.2 Error Amplification

An initial error may become increasingly influential.

Consider a research agent that retrieves an irrelevant article. The error initially affects only source selection. A summarization agent then converts it into a structured finding. A synthesis agent combines it with other evidence. Finally, the writing agent uses the synthesized finding to produce a strong conclusion.

Thus, the original retrieval error becomes a research-level claim.

4.3 Error Masking

Language models are capable of producing fluent and coherent text. This creates a serious reliability problem: downstream generation can make an incorrect intermediate result appear credible.

The final document may therefore contain:

- correct grammar;
- appropriate academic structure;
- realistic citations;
- technically sophisticated terminology;

while still containing unsupported conclusions.

Surface quality is therefore not equivalent to epistemic reliability.

4.4 Correlated Errors

Multiple agents do not necessarily provide independent verification.

If several agents use the same model, same source, same intermediate summary, or same erroneous assumption, they may all reach the same incorrect conclusion.

Consequently:

Multiple agents ≠ independent evidence.

4.5 Feedback-Loop Errors

Agentic systems frequently use feedback to update subsequent decisions. Correct feedback can reduce errors, but incorrect feedback can reinforce them.

Reflexion illustrates how linguistic feedback can improve future agent behavior [2]. However, the effectiveness of such a mechanism depends on whether the feedback accurately identifies the underlying failure.

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5. Error Propagation in Automated Research

Research is especially sensitive to propagated errors because scientific knowledge is cumulative.

Consider an automated literature-review workflow.

Stage 1: Question Interpretation

The agent incorrectly interprets a key term in the research question.

Stage 2: Search

The search agent constructs queries based on the incorrect interpretation.

Stage 3: Source Selection

The screening agent selects sources that fit the incorrect interpretation.

Stage 4: Evidence Extraction

The system extracts legitimate findings from those sources.

Stage 5: Synthesis

The findings are combined into a coherent argument.

Stage 6: Citation

The sources are attached to individual claims.

Stage 7: Report Generation

The final agent produces a polished research paper.

Every source in the final paper could be genuine, yet the paper could still answer the wrong research question.

This demonstrates why **citation authenticity is necessary but not sufficient for research reliability**.

Research automation therefore needs to evaluate the complete chain:

Question → Source → Evidence → Interpretation → Claim → Conclusion

An error anywhere in this chain can compromise the final research product.

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6. Current Approaches and Developments

6.1 External Grounding

External grounding allows agents to obtain information from sources outside the language model. ReAct demonstrated that interaction with external information can help reduce hallucination and error propagation in reasoning tasks [1].

For research automation, grounding can involve:

- scholarly databases;
- publisher websites;
- DOI registries;
- institutional repositories;
- primary datasets;
- official documentation.

However, grounding does not guarantee correctness. A system can retrieve a real source and still interpret it incorrectly.

6.2 Tool-Augmented Research

Tools can provide capabilities for searching, computation, document processing, and data analysis. They can therefore reduce the need for the model to perform tasks solely through text generation.

The key reliability requirement is to preserve the connection between the tool result and the final claim.

For example, if an agent calculates a statistical value using a tool, the final report should retain sufficient provenance to establish:

Input Data → Calculation → Result → Claim

6.3 Reflection and Iterative Correction

Reflection allows an agent to inspect previous outcomes and attempt to improve them. Reflexion demonstrated the value of linguistic feedback and episodic memory for improving agent performance [2].

For research automation, reflection could be used to ask:

- Did the retrieved source actually answer the research question?
- Does the extracted evidence support the summary?
- Does the conclusion follow from the evidence?
- Are any claims unsupported?

However, reflection should preferably be combined with external evidence rather than relying solely on the model's own judgment.

6.4 Independent Verification

A research workflow can separate generation from verification.

For example:

Research Agent → Verification Agent → Final Synthesis

The verification agent can independently check:

- source existence;
- bibliographic metadata;
- numerical results;
- evidence passages;
- claim-source relationships.

Independent verification is particularly important for high-impact claims.

6.5 Provenance Tracking

A provenance-aware research system should record the origin of each important claim.

A provenance chain could contain:

1. research question;
2. search query;
3. source identifier;
4. relevant passage;
5. extracted evidence;
6. transformation or reasoning step;
7. final claim.

This makes it easier to locate the stage at which an error entered the workflow.

6.6 Human-in-the-Loop Systems

Human oversight can be introduced at high-risk workflow boundaries.

Instead of reviewing every generated word, researchers can inspect:

- the research plan;
- important source selections;
- controversial interpretations;
- high-impact conclusions;
- unresolved uncertainty.

Human oversight is particularly valuable when errors have consequences that cannot be reliably evaluated automatically.

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7. Research Challenges and Limitations

7.1 Lack of Standard Error-Propagation Metrics

Current agent benchmarks provide useful information about task success and agent performance, but standardized measures specifically designed to quantify error propagation across research workflows remain limited.

Possible metrics include:

- **Stage Error Rate:** errors introduced at each stage.
- **Propagation Rate:** proportion of stage errors that affect later stages.
- **Amplification Factor:** change in impact between the original error and final output.
- **Recovery Rate:** percentage of errors successfully detected and corrected.
- **Undetected Error Rate:** errors that survive until final output.

- **Claim-Support Accuracy:** proportion of claims correctly supported by their cited evidence.

7.2 Causal Attribution

When a final research report is wrong, identifying the original cause can be difficult. The failure may originate from planning, retrieval, source selection, extraction, reasoning, memory, or synthesis.

Consequently, agentic systems need better observability and execution logs.

7.3 Long-Horizon Reliability

AgentBench found that long-term reasoning, decision-making, and instruction following remain important obstacles for agent systems [3].

Research workflows can be considerably longer than ordinary question-answering tasks. This increases the importance of state management, intermediate verification, and recovery mechanisms.

7.4 Verification Dependence

A verification system may itself be based on an LLM and can therefore reproduce the same biases or reasoning failures as the system being evaluated.

Verification should therefore incorporate independent evidence whenever possible.

7.5 Reproducibility

Web-based research environments are dynamic. Search results, webpages, APIs, and databases can change over time.

A reproducible research agent should preserve:

- search queries;
- source identifiers;
- retrieval dates;
- retrieved evidence;
- model/version information;
- tool outputs;
- important intermediate decisions.

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8. Research Gaps and Future Directions

8.1 Workflow-Level Benchmarks

Future benchmarks should evaluate complete research workflows rather than isolated agent capabilities. Controlled experiments could introduce known errors at specific stages and measure whether agents detect and correct them.

8.2 Causal Error Tracing

Future systems should identify where an error originated and determine which downstream outputs were affected.

This could produce an **error lineage graph**:

Initial Error → Affected Intermediate Results → Affected Claims → Affected Conclusions

Such graphs would make debugging complex research agents more systematic.

8.3 Independent Multi-Agent Verification

Future architectures should distinguish between ordinary multi-agent collaboration and genuine independent verification.

Independent agents should ideally have:

- separate prompts;
- separate evidence retrieval;
- access to primary sources;
- different reasoning paths;
- explicit disagreement mechanisms.

8.4 Evidence-Centered Generation

Instead of allowing the writing agent to generate claims freely, future systems could require every factual claim to be linked to an evidence record before it is included in the final report.

This would change the workflow from:

Generate → Cite

to:

Retrieve → Verify → Extract Evidence → Construct Claim → Cite

8.5 Uncertainty-Aware Research Agents

Agents should explicitly represent uncertainty instead of presenting every conclusion with the same level of confidence.

Potential categories include:

- verified;
- strongly supported;
- partially supported;
- uncertain;
- unsupported.

8.6 Human-AI Collaborative Research

The most reliable near-term architecture may not be complete autonomy. Instead, research agents could automate repetitive tasks while humans retain responsibility for interpretation and important decisions.

The objective should therefore be **controlled autonomy** rather than unrestricted autonomy.

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9. Conclusion

Multi-step agentic AI workflows have significant potential for research automation. They can support planning, information retrieval, evidence extraction, reasoning, synthesis, and report generation. Approaches such as ReAct demonstrate the value of combining reasoning with external actions, while reflection-based methods such as Reflexion demonstrate the potential of feedback and memory for improving agent performance [1,2]. AgentBench further demonstrates that evaluating LLMs as agents requires attention to long-term reasoning, decision-making, and interaction rather than only isolated responses [3].

At the same time, multi-step workflows introduce new reliability challenges. An error introduced during planning, retrieval, source selection, evidence extraction, or reasoning can become an input to later stages. Subsequent agents may transform that error into increasingly polished and consequential outputs. The resulting research report may therefore appear credible while being based on an incorrect premise.

Reliable research automation consequently requires more than powerful language models. It requires **continuous validation, external grounding, independent verification, provenance tracking, uncertainty representation, error tracing, and appropriate human oversight**.

The most important future research direction is to move from evaluating whether an agent can complete a task toward evaluating **how reliably an entire workflow produces evidence-supported conclusions**. Research systems should be able to identify where errors enter, determine how those errors propagate, prevent them from contaminating later stages, and recover when failures occur.

Ultimately, trustworthy agentic research automation should be evaluated not by the fluency of its final report but by the integrity of the complete chain connecting the **research question, evidence, reasoning, citations, and conclusions**.

References

1. **Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y.** (2023). *ReAct: Synergizing Reasoning and Acting in Language Models*. International Conference on Learning Representations (ICLR 2023). arXiv:2210.03629. DOI: 10.48550/arXiv.2210.03629.

2. **Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., & Yao, S.** (2023). *Reflexion: Language Agents with Verbal Reinforcement Learning*. arXiv:2303.11366. DOI: 10.48550/arXiv.2303.11366.

3. **Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Huang, M., Dong, Y., & Tang, J.** (2024). *AgentBench: Evaluating LLMs as Agents*. International Conference on Learning Representations (ICLR 2024). arXiv:2308.03688. DOI: 10.48550/arXiv.2308.03688.

4. **Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., & Scialom, T.** (2023). *Toolformer: Language Models Can Teach Themselves to Use Tools*. arXiv:2302.04761. DOI: 10.48550/arXiv.2302.04761.

5. **Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., & Anandkumar, A.** (2023). *Voyager: An Open-Ended Embodied Agent with Large Language Models*. arXiv:2305.16291. DOI: 10.48550/arXiv.2305.16291.

6. **Dongre, V., Yang, X., Acikgoz, E. C., Dey, S., Tur, G., & Hakkani-Tür, D.** (2024). *ReSpAct: Harmonizing Reasoning, Speaking, and Acting Towards Building Large Language Model-Based Conversational AI Agents*. arXiv:2411.00927. DOI: 10.48550/arXiv.2411.00927.

7. **Zhou, S., Xu, F. F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., et al.** (2024). *WebArena: A Realistic Web Environment for Building Autonomous Agents*. International Conference on Learning Representations (ICLR 2024).

8. **Deng, X., Gu, Y., Zheng, B., Chen, S., Stevens, S., Wang, B., Sun, H., & Su, Y.** (2023). *Mind2Web: Towards a Generalist Agent for the Web*. Advances in Neural Information Processing Systems (NeurIPS 2023).

9. **Qin, Y., Liang, S., Ye, Y., Zhu, K., Yan, L., Lu, Y., Cong, X., Tang, X., Qian, B., Zhao, S., Xie, R., Zhou, J., Gerstein, M., Liu, Z., & Sun, M.** (2023). *ToolLLM: Facilitating Large Language Models to Master 16,000+ Real-World APIs*. arXiv:2307.16789. DOI: 10.48550/arXiv.2307.16789.

10. **Yao, S., Yu, D., Zhao, J., Shafran, I., Narasimhan, K., & Cao, Y.** (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*. Advances in Neural Information Processing Systems (NeurIPS 2023). arXiv:2305.10601. DOI: 10.48550/arXiv.2305.10601.